

Space Insertion

Imagine that you are the chief algorithm officer at a large old company that has a warehouse full of handwritten paper records. The chief record digitization officer has been conducting a long-term project of scanning those paper records to image files, and then using text recognition technology to convert the images to text that can be easily stored in the company's databases. However, some of the handwriting on the records is awful and the text recognition technology is imperfect, so the final digital text that is extracted from a paper record is sometimes incorrect. You've been given only the digitized text and you're asked to find a way to correct the mistakes without referring to the paper originals.

Suppose that you read the first digitized sentence into Python and find that it's a quote from G. K. Chesterton: "The one perfectly divine thing, the one glimpse of God's paradise given on earth, is to fight a losing battle—and not lose it." You take this imperfectly digitized text and store it in a variable called `text`:

```
text = "The oneperfectly divine thing, the oneglimpse of God's paradisegiven on  
earth, is to fight a losingbattle - and notlose it."
```

You'll notice that this text is in English, and while the spelling of each word is correct, there are missing spaces throughout: `oneperfectly` should actually be `one perfectly`, `paradisegiven` should be `paradise given`, and so on. (Missing a space is uncommon for humans, but text recognition technology often makes this kind of mistake.) In order to do your job, you'll have to insert spaces at the appropriate spots in this text. For a fluent English speaker, this task may not seem difficult to do manually. However, imagine that you need to do it quickly for millions of scanned pages—you will obviously need to write an algorithm that can do it for you.

Defining a Word List and Finding Words

The first thing we will do is teach our algorithm some English words. This isn't very hard: we can define a list called `word_list` and populate it with words. Let's start with just a few words:

```
word_list = ['The', 'one', 'perfectly', 'divine']
```
